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EXECUTIVE SUMMARY

*Innovative Automobile engineer with 4+ years of experience in automotive design, manufacturing, and testing. Led the implementation of advanced CAD software techniques using SolidWorks and CATIA for detailed modelling; produced over 200 unique parts within strict timelines, enhancing manufacturing readiness for prototype testing especially in **Electric** and **Hydrogen Fuel cell** vehicles*

Eager to be part of your incredible growth journey, contributing to the advancement of vehicle chassis design and electric powertrain systems. Dedicated to delivering innovative solutions that enhance performance, efficiency, and sustainability, while aligning with the company's vision for excellence.

INSTITUTIONAL PROJECTS

TEAM CAPTAIN & MECHCANICAL LEAD	July 2023 – February 2025
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Team ReNew - Hydrogen Fuel Cell vehicle	Coimbatore, India
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Contributed to an **International team** focused on building a **hydrogen fuel cell vehicle**, with an emphasis on efficiency and sustainability.

- **Leadership:** Led a cross-functional team in designing and constructing a hydrogen fuel cell car for the Shell Eco Marathon, resulting in a 20% reduction in carbon emissions compared to traditional vehicles.
- **Vehicle Development:** Served as the mechanical specialist, working on the chassis, outer panels, and materials engineering, while guiding the design of steering, powertrain, braking, and safety systems.
- **Material Engineering:** Developed 20 custom parts using advanced composite materials (*carbon fibre, PVC foam, aluminium*) to enhance vehicle performance and sustainability.
- **Aerodynamics Optimization:** Boosted overall efficiency by 25% through streamlining design, aerodynamics, and testing processes.
- **Team Management:** Managed an 8-member team, overseeing three major domains, ensuring collaboration and meeting project milestones.
- **Hands-On Expertise:** Enjoyed hands-on experience working with carbon fibres and upcycled PVC foam, particularly in pattern and Mold making, contributing to sustainable design solutions

TEAM CAPTAIN & MECHCANICAL SPECIALIST	December 2021 – July 2023
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Team Zeall Racing - Electric and IC Go-Karts	Coimbatore, India
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- **Vehicle Development:** Implemented advanced testing protocols for both IC and EV go-karts, generating valuable data that informed design improvements, resulting in a 40% reduction in time for new model development.
- **Engine Tuning:** Led advanced engine tuning for optimized power delivery, enhancing performance and efficiency.
- **Battery Pack Design:** Designed custom-made battery packs focused on efficiency and durability, contributing to improved vehicle performance and longevity.
- **Motor Integration:** Integrated a 3-kW motor for high-performance electric propulsion, optimizing power output for the EV go-kart.
- **Team Management:** Managed a team of 60 students, overseeing the development of two go-karts (IC and electric) and guiding them through the design, manufacturing, and testing processes

TEAM CAPTAIN & MECHCANICAL LEAD	December 2021-July2023
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Moto Student India – Team Moto KCT	Coimbatore, India
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Leadership & Team Management: Led an 80-member multidisciplinary team, overseeing project planning, resource allocation, and cross-functional coordination across design, fabrication, and testing phases.

- **Vehicle Development:** Spearheaded the design and development of a **250cc IC** sports bike and a **6kW** electric sports bike, achieving a **top speed of 120 km/h**.
- **Hands-on Expertise:** Actively involved in Designing, manufacturing and assembling **chassis, triple clamp, sprockets, swing-arm, outer panels**, and complete custom EV powertrain from motor selection to final delivery.
- **Material Engineering:** Applied lightweight materials and structural design principles to optimize strength-to-weight ratio, improving performance and handling. Overall weight – 120kg
- **Optimization:** Focused on reducing weight and improving powertrain efficiency through custom design and performance validation. Overall weight – 120kg
- **Achievements:** Secured Overall Championship (twice), along with Cost Evaluation, Design, and Endurance Awards. KTM Raider Training Program certified.
- **National Champion – Season 1 & 2 | FMAE Moto Student India**

FOUNDER & PROJECT MANAGER

Vehicle Development Team @ Research Cell, Kumaraguru Institutions (2021–2024)

- **Leadership & Team Management:** Founded and led the Vehicle Development Unit, guiding **15+ renewable-energy-focused projects** with a team of **55 students**.
- **Project Management:** Oversaw both operational and administrative functions, ensuring timely delivery and quality execution of research and prototyping initiatives.
- **Technical Strategy:** Promoted sustainable design approaches with a strong focus on real-world problem-solving using electric mobility.
- **Recognition:** Played a pivotal role in initiating NEBC, earning accolades from SAE India and expanding national reach to **700+ colleges**.

KEY PROJECTS

RIG-Bike (Intra-Campus Mobility EV, 2021–2024)

- Led a 10-member team to design and build a **lightweight electric bike** aimed at improving faculty mobility across a large campus.
- Integrated a **250W motor with a 1kWh battery**, achieving **75 km range per charge** with a focus on ease of use and low maintenance.
- Earned positive feedback from users and received multiple awards at SAE events hosted by **SRM University**.

NEBC – National Electric Bike Challenge (2022–2023)

- Initiated and coordinated NEBC for two consecutive years, transforming it into a **prestigious national-level competition**.
- Authored the **technical rulebook**, distributed across **700+ institutions**, fostering innovation in electric bike development among student teams nationwide.
- Contributed to technical scrutiny, event operations, and participant mentoring, reinforcing NEBC’s position in India’s academic motorsport scene.

TRACK MARSHAL & EVENT CONTRIBUTOR

Various Motorsport Events & Championships | 2021 – 2025

- **Track Marshalling:** Provided on-ground track safety, recovery support, and race logistics for multiple national and institutional motorsport events including:
 - **MotoStudent India** (KTM & FMAE collaboration)
 - **National Electric Bike Challenge (NEBC)**
 - **Formula Kart Design Challenge (FKDC)**
 - **Go-Kart Design Challenge (GKDC)**
 - **MECO Meritus Cup** and other races at **Kari Motor Speedway** and **CoAST High Performance Tracks**
- **Content Creation:** Served as a **content creator** for the **MRF FMSCI National Dirt Track Championship**, contributing to digital media, race coverage, and promotional campaigns.
- Developed a deep understanding of **race management**, **motorsport safety protocols**, and **event dynamics** from both operational and media perspectives.

PERSONAL PROJECT	2021 – 2024
Impact Resistance System	<i>Coimbatore, India</i>
• Developed impact resistance system for problem-based learning class that reduces accident impact by adjusting the seats. This has drastically reduced the impact of collision to 20%. • This was presented as a paper presentation for our Institution and completed the course. • Registered for patent.	
Air Bag Integrated gloves	
• Ideated and developed a concept for airbag-integrated gloves aimed at enhancing safety for motorcyclists and drivers. • Delivered the project as part of the Kumaraguru Hackathon competition, showcasing innovative safety solutions. • Proposed lightweight, durable materials and user-centric design for comfort and practicality in real-world scenarios.	
Auxiliary Battery Pack development	
• Designed a lightweight 12V, 4.2Ah lithium-ion battery pack with an integrated monitoring system for a hydrogen fuel cell-powered velomobile. • Focused on optimizing the design for efficient jump-starting of the fuel cell system while ensuring durability and safety. • Conducted extensive research and implemented advanced techniques to achieve a compact, lightweight, and high-performance energy storage solution.	

EV DESIGN – MECHANICAL INTERN

Loop Charging Module (LCM), a subsidiary of Eloop Charging and Solutions

June 2025 – Present

3-month Internship

- **Vehicle Efficiency R&D:** Contributing to the development of an on-the-go energy efficiency enhancement system for electric vehicles by integrating a custom-built LCM (Loop Charging Module) component.
- **CAD Design Integration:** Working as a **CAD designer** for retrofitting the LCM prototype into an existing EV platform, ensuring packaging, compatibility, and structural feasibility.
- **Mechanical Development:** Supporting prototype adaptation and mounting solutions to improve system efficiency without compromising vehicle dynamics or design standards.
- **Innovation Focus:** Part of a team developing novel solutions for enhancing **real-time EV performance and range**.

SKILLS

Engineering Tools	Level	Domains
SOLIDWORKS	Expert	Weldments -Solid modelling- Surface modelling - Topology optimisation
ANSYS WORKBENCH	Expert	Static Structural - Fluid Flow Fluent - ACP (Pre, Post) -Meshing
SOLIDWORKS VISUALIZE	expert	
ABACUS SIMULATION	Beginner	
Siemens NX	Beginner	
Creo	Beginner	
Fusion 360	Beginner	
Air shaper	Beginner	
MATLAB	Beginner	

Extra skills	Soft Skills
• ADOBE ILLUSTRATOR	• Leadership
• ADOBE LIGHTROOM	• Communication
• ADOBE PHOTOSHOP	• Problem-Solving
• BLENDER	• Teamwork
• DAVINICI RESOLVE	• Time Management
• PREMIRE PRO	• Creativity
• AFTER EFFECTS	• Attention to Detail

Awards from Kumaraguru Institution

- Certificate Of Recognition 2022
- Certificate Of Recognition 2023
- Certificate Of Recognition 2024
- **Achiever’s Award 2024 – Mechanical Techspert**
- **Garage Techspert’s Award 2024 - Technical performance**
- **Impact Resistance System - Certified from Kumaraguru Institution's Problem Based Learning Platforms.**

Awards Across Various International and National Events

- Best Captain award
- Skid pad Award
- Autocross Award
- Endurance winner
- Safety Award
- Go Green, Innovation,
- Lightweight vehicle Award
- Business Plan Award
- Winner of overall championships for 2 consecutive years in Moto Student, along with the Design Prize and Best Cost Prize (2022-2023).

EDUCATION

Bachelor of Engineering – Automobile Engineering

Anna University, India (Kumaraguru College of Technology)

2021-2025

AREA OF INTRESTS

Design & Engineering, Materials & Manufacturing, Powertrain & Performance, Safety & Compliance, Product Engineering, Simulation & Testing, Transmission System, Topology Optimisation, Aerodynamics, Composite materials, Advanced 3d Prototyping, Chassis and vehicle panel designing.